Hypercube Sums via a Dual Kronecker Chain: Sumcheck on Growing Domains, Polynomial-Valued FRI on Shrinking Domains

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Abstract

We study an interactive oracle proof for the Boolean-hypercube sum $H = \sum_{b\in\{0,1\}^m} f(b)$ of a committed multilinear polynomial f. The protocol runs two synchronized chains on the same challenges $r_1, \ldots, r_m$: a forward Kronecker–Sumcheck chain on growing multiplicative domains $D_0 \subset \cdots \subset D_m$, and a backward polynomial-valued (symbolic) FRI chain on shrinking domains, whose leaves are polynomials in a formal variable X. Both chains terminate at $f(r_1 X^{2^{m-1}}, \ldots, r_m X)$. We prove completeness for every m, show that in the exact-identity model both chains are deterministic bijections, give two explicit attacks on under-specified variants (one accepting a false sum with probability 1), and prove a soundness theorem for the repaired protocol with the explicit bound $m/|K| + 2B(2^m - 1)/|K| + (1-\theta)^q$, $\theta = (1 - 1/B)/2$. For Goldilocks^3, $m = 20$, $B = 4$, this gives $q = 189$ queries for 128-bit interactive soundness. The binding argument for the polynomial-valued chain is proximity-gap free: it uses only injectivity of the formal fold. Its price is a proof of size $\Theta(2^m)$. Specializing $X := 1$ collapses the protocol to classical sumcheck interleaved with scalar FRI; for this succinct variant we prove evaluation binding up to the unique-decoding radius, from a Berlekamp–Welch correlated-agreement lemma for Reed–Solomon lines over $K(Z)$, with bound $m/|K| + (n^2/3 + 4n + m)/|K| + (1-\theta)^q$, again $q = 189$. The two variants are validated numerically by an accompanying Rust implementation.

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1 Setting and notation

Fields. $k = \mathbb{F}_p$ with $p = 2^{64} - 2^{32} + 1$ (Goldilocks), and $K = \mathbb{F}_{p^3}$ a fixed cubic extension. All challenges are sampled uniformly from K.

Parameters. $m \geq 1$ is the number of variables, $L = 2^m$, $B \geq 2$ is the blowup (a power of two), $n = BL$. We require $n \mid p-1$. The rate is $\rho = 1/B$ and the unique-decoding threshold is

$$\theta := \frac{1-\rho}{2}.$$

Domains. For $0 \leq i \leq m$ let $D_i = \mu_{B2^i} \subset k^\times$ be the group of $B2^i$-th roots of unity, so $|D_i| = B2^i$ and $x \mapsto x^2$ maps D_i two-to-one onto D_{i-1}. The shrinking domains are $E_j := D_{m-j}$. The fibers of the top domain over D_0 are

$$\Phi_z := \{y \in D_m : \ y^L = z\}, \qquad z \in D_0, \qquad |\Phi_z| = L.$$

Index conventions. For $S \subseteq [m]$ put

$$j(S) := \sum_{k \in S} 2^{k-1}, \qquad j'(S) := \sum_{k \in S} 2^{m-k}.$$

Both $S \mapsto j(S)$ and $S \mapsto j'(S)$ are bijections $2^{[m]} \to \{0, \ldots, L-1\}$. For an integer $0 \leq t < L$, t_k denotes the bit of weight 2^{k-1}, so $t = \sum_k t_k 2^{k-1}$. For $1 \leq i \leq m$ and $0 \leq t < 2^i$ let $\mathrm{rev}_i(t) := \sum_{k=1}^{i} t_k 2^{i-k}$.

Polynomials. The multilinear $f = \sum_{S \subseteq [m]} c_S X^S$ (with $X^S = \prod_{k \in S} X_k$) has Kronecker encoding

$$F(Y) := f(Y, Y^2, \ldots, Y^{2^{m-1}}) = \sum_S c_S Y^{j(S)}, \qquad \deg F < L.$$

This is a K-linear bijection between multilinear polynomials and $K[Y]_{<L}$; we write $f = \kappa^{-1}(F)$. For $r = (r_1, \ldots, r_m) \in K^m$ define

$$\beta_k(X) := r_k X^{2^{m-k}}, \qquad f(\beta(X)) := f(\beta_1(X), \ldots, \beta_m(X)) = \sum_S c_S \Big(\prod_{k \in S} r_k \Big) X^{j'(S)}.$$

For $0 \le i \le m$ let $A_i(X_1, \dots, X_i) := \sum_{b \in \{0,1\}^{m-i}} f(X_1, \dots, X_i, b)$, so $A_m = f$ and $A_0 = \sum_b f(b)$.

Even/odd split. Every $P \in K[X]$ is uniquely $P(X) = E_P(X^2) + X\, O_P(X^2)$.

Codes. $\mathrm{RS} := \mathrm{RS}[K, D_m, L] = \{F|_{D_m} : F \in K[Y]_{<L}\}$. Relative Hamming distance is $\mathrm{d}(\cdot,\cdot)$. The minimum relative distance of RS is $\lambda = 1 - \rho + 1/n > 2\theta$.

Remark 1.1 (The target is an evaluation)**.** Since $\sum_b \prod_{k \in S} b_k = 2^{m-|S|}$,

$$\sum_{b \in \{0,1\}^m} f(b) = \sum_S c_S\, 2^{m-|S|} = 2^m f\big(\tfrac{1}{2}, \dots, \tfrac{1}{2}\big).$$

A hypercube sum of a single committed multilinear polynomial is therefore an evaluation claim at the point $(\frac{1}{2}, \dots, \frac{1}{2})$. This identity explains the structure found below: both chains are re-encodings of the coefficient vector $(c_S)_S$, and the sum is a fixed linear functional of it.

2 The protocol

2.1 Chain A: growing-domain Kronecker–Sumcheck

Oracles: $P_0 := H$ (a scalar); for $1 \le i \le m$, tables $U_i, V_i : D_{i-1} \to K$ and $P_i : D_i \to K$.

Round i: the prover commits to (U_i, V_i); the verifier samples r_i; the prover commits to P_i. The intended relations are

$$P_{i-1}(Y) = 2U_i(Y) + V_i(Y), \qquad P_i(X) = U_i(X^2) + r_i X\, V_i(X^2).$$

Path check. For a point $x_m \in D_m$ let $x_{i-1} := x_i^2$. The verifier opens, for $i = 1, \dots, m$:

$$\begin{aligned} &(\mathrm{a}_i) \quad P_{i-1}(x_{i-1}) \stackrel{?}{=} 2U_i(x_{i-1}) + V_i(x_{i-1}) \qquad (P_0 := H), \\ &(\mathrm{b}_i) \quad P_i(x_i) \stackrel{?}{=} U_i(x_{i-1}) + r_i x_i\, V_i(x_{i-1}). \end{aligned}$$

2.2 Chain B: shrinking-domain polynomial-valued FRI

Leaf format. A level-i leaf ($0 \le i \le m$) sits at $z \in E_i = D_{m-i}$ and is an element

$$s_i(z, \cdot) \in K[W]_{<2^i}, \qquad \text{representing } S_i(z, X) := s_i(z, X^{2^{m-i}}).$$

It is committed as its canonical vector of 2^i coefficients. Level 0 is the committed table $s_0(y) \in K$, $y \in D_m$ (the oracle C_f).

Fold. For $1 \le i \le m$, $y \in E_{i-1}$, and $a, b \in K[W]_{<2^{i-1}}$ put $e := \frac{a+b}{2}$, $o := \frac{a-b}{2y}$ and

$$\mathrm{Fold}_{i,y}(a, b)(V) := e(V^2) + r_i V\, o(V^2) \ \in K[V]_{<2^i}.$$

With $V = X^{2^{m-i}}$ and $W = V^2 = X^{2^{m-i+1}}$, this is exactly

$$S_i(y^2, X) = \frac{S_{i-1}(y, X) + S_{i-1}(-y, X)}{2} + r_i X^{2^{m-i}}\, \frac{S_{i-1}(y, X) - S_{i-1}(-y, X)}{2y}.$$

In round i, after r_i, the prover commits to the level-i table $z \mapsto s_i(z, \cdot)$ on E_i.

Lemma 2.1 (Injectivity of the fold)**.** *If $r_i \neq 0$ and $y \neq 0$, then $\mathrm{Fold}_{i,y} : K[W]^2_{<2^{i-1}} \to K[V]_{<2^i}$ is a linear bijection.*

Proof. $e(V^2)$ is supported on even powers and $r_i V o(V^2)$ on odd powers, so the image determines (e, o); then $a = e + yo$, $b = e - yo$. Dimensions agree ($2 \cdot 2^{i-1} = 2^i$). □

Chain-B query. For $y_0 \in D_m$ let $y_i := y_{i-1}^2 \in E_i$. The verifier opens $s_i(\pm y_i)$ and $s_{i+1}(y_{i+1})$ for $0 \le i < m$ (pairs $\{y, -y\}$ are grouped in one Merkle leaf), checks the leaf format, and checks $s_{i+1}(y_{i+1}) = \mathrm{Fold}_{i+1,y_i}\big(s_i(y_i), s_i(-y_i)\big)$.

2.3 The repaired protocol

The three items marked (F1)–(F3) are not optional; Section 5 shows what fails without (F1) and why (F2) is load-bearing.

Protocol 2.2 (Π_{dual})**.**

1. Prover sends C_f (oracle $s_0 : D_m \to K$) and H.
2. For $i = 1, \dots, m$: prover commits (U_i, V_i) on D_{i-1}; verifier sends $r_i \leftarrow K$; prover commits P_i on D_i and the chain-B level s_i on E_i.
3. Query phase (all commitments fixed):
   - (F1) Open all B terminal leaves $s_m(z, \cdot)$, $z \in D_0$, in full; reject unless they are equal. Call the common value $T \in K[X]_{<L}$.
   - (F2) Every opened leaf must be a coefficient vector of length exactly 2^i at level i.
   - (F3) Run q chain-B queries from independent uniform $y_0^{(1)}, \dots, y_0^{(q)} \in D_m$.
   - (A) Run one chain-A path check from a point $x_m \in D_m$ (any choice, e.g. $x_m = y_0^{(1)}$), and check $P_m(x_m) = T(x_m)$.
4. Accept iff all checks pass.

3 Completeness

Lemma 3.1 (Honest chain A)**.** *Define, for* $1 \le i \le m$,

$$Q_i(Y, Z) := A_i\big(r_1 Y^{2^{i-2}}, r_2 Y^{2^{i-3}}, \dots, r_{i-1} Y,\ Z\big), \qquad Q_1(Y, Z) := A_1(Z),$$

write $Q_i = U_i(Y) + Z\, V_i(Y)$*, and set* $P_i(X) := A_i(r_1 X^{2^{i-1}}, \dots, r_{i-1} X^2, r_i X)$*,* $P_0 = A_0$*. Then for all* i*:*

(i) Q_i *is affine in* Z*, and* $\deg U_i, \deg V_i \le 2^{i-1} - 1$*;*

(ii) $P_{i-1}(Y) = 2U_i(Y) + V_i(Y)$*;*

(iii) $P_i(X) = U_i(X^2) + r_i X\, V_i(X^2)$*;*

(iv) $P_m(X) = f(\beta(X))$*.*

Proof. (i) A_i is multilinear; the Y-degree is at most $\sum_{t=0}^{i-2} 2^t = 2^{i-1} - 1$. (ii) $Q_i(Y, 0) + Q_i(Y, 1) = A_{i-1}(r_1 Y^{2^{i-2}}, \dots, r_{i-1} Y) = P_{i-1}(Y)$, using $A_{i-1}(\cdot) = A_i(\cdot, 0) + A_i(\cdot, 1)$. (iii) $Q_i(X^2, r_i X) = A_i(r_1 X^{2^{i-1}}, \dots, r_{i-1} X^2, r_i X) = P_i(X)$. (iv) is (iii)'s definition at $i = m$, since $A_m = f$. □

Lemma 3.2 (Honest chain B)**.** *Let* $F = \sum_{t<L} c_t Y^t$ *and* $s_0 = F|_{D_m}$*, and fold honestly. Then for* $0 \le i \le m$ *and* $z \in E_i$*,*

$$s_i(z, V) = \sum_{t<L} c_t\ z^{\lfloor t/2^i \rfloor} \Big(\prod_{k \le i,\ t_k = 1} r_k \Big)\, V^{\mathrm{rev}_i(t \bmod 2^i)}.$$

In particular $\deg_V s_i < 2^i$*, and* $S_m(z, X) = s_m(z, X) = f(\beta(X))$ *for every* $z \in D_0$*.*

Proof. Induction on i; $i = 0$ is the definition. Assume the formula at level $i-1$ and fix $y \in E_{i-1}$, $z = y^2$. Write $u = \lfloor t/2^{i-1} \rfloor$, so $t_i = u \bmod 2$. Then $\frac{y^u + (-y)^u}{2} = y^u[t_i = 0]$ and $\frac{y^u - (-y)^u}{2y} = y^{u-1}[t_i = 1]$, and in both cases $y^{u-t_i} = z^{\lfloor t/2^i \rfloor}$. The term with $t_i = 0$ goes to $V^{2\mathrm{rev}_{i-1}(t)}$, the term with $t_i = 1$ to $r_i V^{1+2\mathrm{rev}_{i-1}(t)}$, and $\mathrm{rev}_i(t \bmod 2^i) = 2\mathrm{rev}_{i-1}(t \bmod 2^{i-1}) + t_i$. At $i = m$: $\lfloor t/L \rfloor = 0$ and $X^{\mathrm{rev}_m(j(S))} = X^{j'(S)}$. □

Corollary 3.3 (Completeness)**.** *The honest prover is accepted with probability* 1*: all leaves are well formed, all B terminal leaves equal $f(\beta) = P_m$, and every local check is an instance of Lemma 3.1 or 3.2.*

Remark 3.4 (Orientation of the powers)**.** With the tempting definition

$$Q_i(Y, Z) = A_i\big(r_1 Y^{2^{i-1}}, \ldots, r_{i-1} Y^2, Z\big),$$

identity (ii) would require the first of the following, whereas Lemma 3.1 gives the second:

$$P_{i-1}(Y) \stackrel{?}{=} A_{i-1}\big(r_1 Y^{2^{i-1}}, \ldots, r_{i-1} Y^2\big), \qquad P_{i-1}(Y) = A_{i-1}\big(r_1 Y^{2^{i-2}}, \ldots, r_{i-1} Y\big);$$

the two differ for $i \geq 2$ in general. Lemma 3.1 is the consistent orientation.

4 The exact-identity model

Suppose every round relation holds as a polynomial identity, all $r_i \neq 0$, and the leaves have the format (F2).

Proposition 4.1 (Chain A is invertible)**.** *The map $(U_i, V_i) \mapsto P_i$ with $P_{i-1} = 2U_i + V_i$ is inverted by $U_i = E_{P_i}$, $V_i = r_i^{-1} O_{P_i}$, so $P_{i-1} = 2E_{P_i} + r_i^{-1} O_{P_i}$. Iterating,*

$$H = P_0 = \Lambda_r(P_m), \qquad \Lambda_r\Big(\sum_S p_S\, X^{j'(S)}\Big) := \sum_S p_S\, 2^{m-|S|} \prod_{k \in S} r_k^{-1}.$$

If $P_m = f(\beta)$, i.e. $p_S = c_S \prod_{k \in S} r_k$, then $H = \sum_S c_S 2^{m-|S|} = \sum_b f(b)$.

Proof. At round i the lowest power X^1 of P_i carries variable X_i, so the odd part corresponds to $i \in S$; the inverse step multiplies coefficients with $i \notin S$ by 2 and those with $i \in S$ by r_i^{-1}. Compose over $i = m, \ldots, 1$ and use Remark 1.1. □

Proposition 4.2 (Chain B is a bijective re-encoding)**.** *For fixed r with all $r_k \neq 0$, $f \mapsto f(\beta)$ is a bijection from multilinear polynomials onto $K[X]_{<L}$, and by Lemma 2.1 each fiber Φ_z of level* 0 *is mapped bijectively onto the terminal leaf $s_m(z, \cdot)$.*

Remark 4.3. In the exact-identity model, neither chain uses the randomness of r: chain A computes the fixed functional Λ_r, which on honest inputs is the r-independent value $\sum_b f(b)$, and chain B is a deterministic butterfly of F. In particular a prover cannot satisfy the recurrences with an alternative polynomial family, since the recurrences are invertible. All soundness content therefore lies in the passage from sampled checks to identities, which is where the query analysis of Section 6 lives.

5 Attacks on under-specified variants

5.1 Fiber splitting: false sum accepted with probability one

Let Π^- be Π_{dual} with (F1) replaced by: *every check that needs a terminal value at x_m (resp. at the end of a chain-B query) uses the leaf $s_m(x_m^L, \cdot)$ (resp. $s_m(y_m, \cdot)$) of its own fiber.*

Observe that both chains split into B independent subtrees indexed by $z \in D_0$: in chain B, $\pm y$ and y^2 lie over the same z; in chain A, x and x^2 satisfy $x^{2^i} = (x^2)^{2^{i-1}}$.

Proposition 5.1. *For every multilinear $f_\star$ and every $H' \neq \sum_b f_\star(b)$ there is a prover for Π^- whose initial table satisfies $\mathrm{d}(s_0, F_\star) = B/n = 1/L$ (so $F_\star = \kappa(f_\star)$ is its unique decoding) and which is accepted with probability 1 claiming H'.*

Proof. For $z \in D_0$ pick $y_z \in \Phi_z$ and set

$$h_z(Y) := \frac{Y^L - z}{Y - y_z} = \sum_{t=0}^{L-1} y_z^{L-1-t} Y^t.$$

It vanishes on $\Phi_z \setminus \{y_z\}$ and $h_z(y_z) = L y_z^{L-1} \neq 0$. Let $\eta_z := \kappa^{-1}(h_z)$. By Remark 1.1,

$$\sigma_z := \sum_b \eta_z(b) = \sum_{t<L} y_z^{L-1-t} 2^{m-\mathrm{wt}(t)} = y_z^{L-1} \prod_{k=1}^{m} \left(2 + y_z^{-2^{k-1}}\right).$$

For each k at most 2^{k-1} elements of the coset Φ_z satisfy $y^{2^{k-1}} = -\frac{1}{2}$, so at most $L - 1 < |\Phi_z|$ choices are excluded, and we may choose y_z with $\sigma_z \neq 0$. Put $\lambda_z := (H' - H_\star)/\sigma_z$ with $H_\star := \sum_b f_\star(b)$, and

$$s_0(y) := F_\star(y) + \lambda_z h_z(y) \qquad (y \in \Phi_z).$$

This differs from $F_\star$ exactly at the B points y_z. On subtree z the prover runs the honest prover (both chains, Lemmas 3.1 and 3.2) for $g_z := f_\star + \lambda_z \eta_z$, whose Kronecker encoding interpolates s_0 on Φ_z. Every local check holds identically, the leaf $s_m(z, \cdot) = g_z(\beta)$ matches chain A on the same subtree, and $\sum_b g_z(b) = H_\star + \lambda_z \sigma_z = H'$ for every z, consistent with the single scalar $P_0 = H'$. □

Under (F1) this prover is rejected with certainty: $g_z(\beta) = g_{z'}(\beta)$ would force $\lambda_z h_z = \lambda_{z'} h_{z'}$, but h_z vanishes on $\Phi_z \setminus \{y_z\}$ while $h_{z'}(y) = (z - z')/(y - y_{z'}) \neq 0$ there.

5.2 Late commitment of the first round message

Proposition 5.2. *If (U_1, V_1) is committed after r_1, then for any $H' \neq H_\star$ there is a prover that is honest in chain B and passes each independent uniform chain-A path with probability 1/2.*

Proof. For each $y \in D_0$ choose a root $x_y \in D_1$ with $2r_1 x_y \neq 1$ and solve $2U + V = H'$, $U + r_1 x_y V = P_1^\star(x_y)$ (determinant $2r_1 x_y - 1$). Commit $P_1 := P_1^\star$ and play honestly from round 2 on. Only (b_1) can fail, and only when $x_1 = -x_y$; for uniform x_m, $x_1 = x_m^{2^{m-1}}$ is uniform on D_1. □

With the correct order the corresponding error is at most $2B(2^m - 1)/|K|$, about 2^{-169} (Lemma 6.3), so the late variant drops chain A from field-size soundness to 2^{-q_A} for q_A paths.

5.3 Role of (F2) in the proof

The detection argument (Lemma 6.5) walks up a query path and uses injectivity of $\mathrm{Fold}_{k,y}$ on the opened pair. On a query path only the path node is checked against its own children; the sibling is not. Lemma 2.1 gives injectivity only on $K[W]^2_{<2^{k-1}}$, and without the format check a sibling could be an arbitrary polynomial of larger degree, for which $(a, b) \mapsto e(V^2) + r_i V o(V^2)$ has a nontrivial kernel. (F2) costs nothing and also fixes the leaf size at 2^i coefficients, which is what makes the cost bounds of Section 7 hold.

6 Soundness

6.1 The binding game

Definition 6.1 (Reference polynomial)**.** Given the committed oracle $s_0 : D_m \to K$, let $F_\star$ be the unique $F \in K[Y]_{<L}$ with $\mathrm{d}(s_0, F|_{D_m}) < \theta$, if it exists (unique because $2\theta < \lambda$), and $f_\star := \kappa^{-1}(F_\star)$, $H_\star := \sum_b f_\star(b)$.

Definition 6.2 (Game)**.** An unbounded interactive oracle prover $\widetilde{P}$ interacts with the verifier of Protocol 2.2. $\widetilde{P}$ wins if the verifier accepts and either $F_\star$ does not exist or $H \neq H_\star$.

Note that $F_\star$ is a function of s_0 alone, hence fixed before r_1 is sampled.

6.2 Chain A: pathwise sumcheck

Let $U_i^\star, V_i^\star, P_i^\star$ be the honest tables of Lemma 3.1 for $f_\star$; they depend on $f_\star$ and $r_1, \ldots, r_{i-1}$ only (and $P_i^\star$ also on r_i).

Lemma 6.3 (Pathwise sumcheck)**.** *Assume $F_\star$ exists. Let Bad_A be the event that for some i, some $y \in D_{i-1}$ and some x with $x^2 = y$,*

$$(U_i(y), V_i(y)) \neq (U_i^\star(y), V_i^\star(y)) \quad \textit{and} \quad U_i(y) + r_i x V_i(y) = U_i^\star(y) + r_i x V_i^\star(y).$$

Then $\Pr[\mathsf{Bad}_A] \leq \sum_{i=1}^m 2|D_{i-1}|/|K| = 2B(2^m - 1)/|K|$. *Moreover, outside Bad_A, for* every $x_m \in D_m$*: if $H \neq H_\star$ and all checks $(\mathrm{a}_i), (\mathrm{b}_i)$ on the path from x_m pass, then $P_m(x_m) \neq f_\star(\beta(x_m))$.*

Proof. Conditioned on the transcript before r_i, the tables $U_i, V_i, U_i^\star, V_i^\star$ are fixed and r_i is uniform. For fixed (y, x) with $(\Delta U, \Delta V) := (U_i - U_i^\star, V_i - V_i^\star)(y) \neq 0$ and $x \neq 0$, the affine function $r \mapsto \Delta U + r x \Delta V$ has at most one root. A union bound over i, y and the two roots x gives the probability bound.

For the second claim let $\Delta_i := P_i(x_i) - P_i^\star(x_i)$, with $\Delta_0 = H - H_\star \neq 0$. If $\Delta_{i-1} \neq 0$ and (a_i) passes, then $2\Delta U + \Delta V = \Delta_{i-1} \neq 0$ at x_{i-1}, so $(\Delta U, \Delta V) \neq 0$; outside Bad_A the right-hand side of (b_i) differs from $P_i^\star(x_i)$, and since (b_i) passes, $\Delta_i \neq 0$. By induction $\Delta_m \neq 0$, and $P_m^\star = f_\star(\beta)$ by Lemma 3.1(iv). □

Remark 6.4. Lemma 6.3 needs no degree test on P_i, U_i, V_i, and one path suffices for any choice of x_m, including a fixed one. Wrongness propagates under pullback: a wrong value at y forces wrong values at both square roots, so the error pattern of a false claim covers all of D_m, the opposite of FRI's distance decay.

6.3 Chain B: proximity-gap-free tree detection

Lemma 6.5 (Tree detection)**.** *Assume all $r_k \neq 0$, (F1) and (F2) pass with common terminal value T, and let $F' \in K[Y]_{<L}$ be the unique polynomial with $\kappa^{-1}(F')(\beta) = T$ (Proposition 4.2). Then every chain-B query started at y_0 with $s_0(y_0) \neq F'(y_0)$ rejects. Consequently*

$$\Pr\left[\textit{all } q \textit{ chain-B queries accept} \;\middle|\; \textit{commitments}\right] \leq \left(1 - \mathrm{d}(s_0, F')\right)^q.$$

Proof. Let s_i' be the honest butterfly of F' (Lemma 3.2); then $s_m'(z) = T$ for all $z \in D_0$, so $W_m = \emptyset$ where $W_i := \{z \in E_i : s_i(z) \neq s_i'(z)\}$. Let $y_0 \in W_0$ with ancestors $y_1, \ldots, y_m$, and let $k \geq 1$ be minimal with $y_k \notin W_k$. If the check at y_k passed, then

$$\mathrm{Fold}_{k,y_{k-1}}\big(s_{k-1}(y_{k-1}), s_{k-1}(-y_{k-1})\big) = s_k(y_k) = s_k'(y_k) = \mathrm{Fold}_{k,y_{k-1}}\big(s_{k-1}'(y_{k-1}), s_{k-1}'(-y_{k-1})\big),$$

and since the opened leaves have the format (F2), Lemma 2.1 gives $s_{k-1}(y_{k-1}) = s_{k-1}'(y_{k-1})$, contradicting $y_{k-1} \in W_{k-1}$. The queries are uniform and independent after T, hence F', is fixed. □

Lemma 6.6 (Distance to a wrong target)**.** *If $F_\star$ does not exist, or exists and $F' \neq F_\star$, then* $\mathrm{d}(s_0, F') \geq \theta$.

Proof. If $F_\star$ does not exist, $\mathrm{d}(s_0, c) \geq \theta$ for every codeword c. Otherwise $\mathrm{d}(s_0, F') \geq \lambda - \mathrm{d}(s_0, F_\star) > \lambda - \theta > \theta$. □

6.4 Main theorem

Theorem 6.7 (Interactive soundness of Π_{dual})**.** *For every unbounded interactive oracle prover,*

$$\Pr[\mathsf{Win}] \;\leq\; \frac{m}{|K|} + \frac{2B(2^m - 1)}{|K|} + (1-\theta)^q.$$

Proof. Partition the winning event.

(1) Some $r_k = 0$: probability at most $m/|K|$.

(2) All $r_k \neq 0$, and $F_\star$ does not exist or $T \neq f_\star(\beta)$. Then F' is well defined and either $F_\star$ is absent or $F' \neq F_\star$, so by Lemmas 6.5 and 6.6 acceptance has probability at most $(1-\theta)^q$.

(3) All $r_k \neq 0$, $F_\star$ exists, $T = f_\star(\beta)$, and $H \neq H_\star$. The check $P_m(x_m) = T(x_m) = f_\star(\beta(x_m))$ together with the chain-A path checks contradicts Lemma 6.3 unless Bad_A holds. □

6.5 Parameters

For $K = \mathbb{F}_{p^3}$ ($|K| \approx 2^{192}$), $m = 20$, $B = 4$ ($\theta = 3/8$):

Term	Value	$\log_2$
$m/\|K\|$	$20/p^3$	-187.7
$2B(2^m-1)/\|K\|$	$8(2^{20}-1)/p^3$	-169.0
$(5/8)^{189}$		-128.16
$(5/8)^{188}$		-127.48

Hence $q = 189$ chain-B queries and a single chain-A path give total interactive error below 2^{-128}, and $q = 189$ is minimal for this bound. This coincides with the unique-decoding query count of the line-binding symbolic-FRI protocol, as expected: both are governed by $(1-\theta)^q$.

Remark 6.8 (Compilation)**.** Under the BCS transform with Merkle commitments in the random-oracle model, add $\varepsilon_{\mathrm{Merkle}}$, the probability that the prover finds a collision (at most $Q^2/2^{\lambda_h}$ for Q queries and λ_h-bit digests). For the Fiat–Shamir NIZK, the query phase is a single round with error $(1-\theta)^q$ per attempt, so a prover with Q hash queries gains a factor of about Q; compensate with grinding or with $q \geq (128 + \log_2 Q - g)/\log_2(8/5)$ for g bits of proof-of-work.

Remark 6.9 (Beyond unique decoding)**.** Lemma 6.5 gives per-query detection $\mathrm{d}(s_0, F')$ for the target F' chosen by the prover. Above θ the target may be any codeword of the list around s_0, selected adaptively after r, so the winning condition must change to a list statement; the deterministic chain alone cannot bind to a single $f_\star$ there.

7 Costs

All counts are in elements of K (24 bytes); $m = 20$, $B = 4$, $q = 189$.

Prover.

- Chain B: level i has $B2^{m-i}$ leaves of 2^i coefficients, i.e. BL elements per level, $mBL \approx 8.39 \cdot 10^7$ elements (≈ 2.0 GB) in total, computed in $O(mBL)$ field operations.
- Chain A: $\sum_{i=1}^{m} 2B2^i \approx 4BL \approx 1.68 \cdot 10^7$ elements; the coefficients of all A_i come from $O(L)$ partial sums, then one NTT per table.

Proof. At level $i < m$ the opened data is at most $\min(q, B2^{m-i-1}) \cdot 2^{i+1}$ elements.

Levels	Opened elements (upper bound)
0–13	$189\,(2^{15} - 2) \approx 6.19 \cdot 10^6$
14–19 (whole levels)	$6 \cdot BL \approx 2.52 \cdot 10^7$
terminal leaves (F1)	$BL \approx 4.19 \cdot 10^6$
chain A (one path)	$3m = 60$
total	$\approx 3.56 \cdot 10^7$ (≈ 0.85 GB)

Verifier. $\Theta(L)$: reading T, evaluating $T(x_m)$, and fold checks whose cost equals the opened data. Once T is read, the verifier could compute $H = \Lambda_r(T)$ directly, so in Π_{dual} chain A is redundant.

8 Collapse: specializing the formal variable

Lemma 8.1 (Specialization)**.** *Fix $x \in k^\times$ and put $\alpha_i := r_i x^{2^{m-i}}$. Then:*

(i) the scalars $g_i(z) := S_i(z, x)$ satisfy the scalar FRI fold

$$g_i(y^2) = \frac{g_{i-1}(y) + g_{i-1}(-y)}{2} + \alpha_i \frac{g_{i-1}(y) - g_{i-1}(-y)}{2y};$$

(ii) along the chain-A path from $x_m = x$, with $(u_i, v_i) := (U_i, V_i)(x_{i-1})$, the checks $(\mathrm{a}_i), (\mathrm{b}_i)$ read

$$P_{i-1}(x_{i-1}) = 2u_i + v_i, \qquad P_i(x_i) = u_i + \alpha_i v_i,$$

which is classical sumcheck with round polynomial $u_i + Zv_i$ and challenge α_i;

(iii) the α_i are independent and uniform in K.

Proof. (i) and (ii) are substitution of $X = x$; note $x_i = x^{2^{m-i}}$ so $r_i x_i = \alpha_i$. (iii) $r_i \mapsto r_i x^{2^{m-i}}$ is a bijection of K. □

The weighting $X^{2^{m-i}}$ is therefore cosmetic once X is specialized, and the same α_i drives both chains. Taking $x = 1$:

Protocol 8.2 (Π_1)**.** Commit $s_0 = F|_{D_m}$ and send H. For $i = 1, \ldots, m$: send $(u_i, v_i) \in K^2$ in the clear, check $2u_i + v_i = c_{i-1}$ ($c_0 = H$), sample r_i, set $c_i = u_i + r_i v_i$, commit the scalar FRI layer g_i on D_{m-i} folded with r_i. Finally check that g_m is constant on D_0 with value c_m, and run q FRI queries.

Π_1 is classical sumcheck for $\sum_b f(b)$ interleaved with scalar FRI on the Kronecker encoding, with shared challenges. The scalar fold is not injective, so Lemma 6.5 does not transfer; the binding argument below replaces injectivity by a correlated-agreement lemma for lines. Two versions are proved: Lemma 8.5 for any linear code (radius $(1 - \rho)/3$) and Lemma 8.7 for Reed–Solomon codes (radius $(1 - \rho)/2$). The rest of the argument uses only the even/odd decomposition of Lemma 8.4 and the minimum distance.

8.1 Pair representation

Throughout, $g_0 := s_0$ and $g_i : E_i \to K$ $(1 \le i \le m)$ are the committed layers of Π_1, $n_i := |E_i| = n/2^i$, $L_i := L/2^i$, and $C_i := \mathrm{RS}[K, E_i, L_i]$, with minimum distance $d_i = n_i - L_i + 1$ and relative distance $\lambda_i = d_i/n_i > 1 - \rho$.

Definition 8.3 (Pairs)**.** For $g : E_{i-1} \to K$ and $z = y^2 \in E_i$ put

$$e_g(z) := \frac{g(y) + g(-y)}{2}, \qquad o_g(z) := \frac{g(y) - g(-y)}{2y}, \qquad \mathrm{Fold}_r(g) := e_g + r\, o_g : E_i \to K.$$

The map $(g(y), g(-y)) \mapsto (e_g(z), o_g(z))$ is invertible. The *pair distance* $\Delta(g, g')$ is the fraction of $z \in E_i$ with $(e_g, o_g)(z) \ne (e_{g'}, o_{g'})(z)$, equivalently $(g(\pm y)) \ne (g'(\pm y))$.

Lemma 8.4 (Even/odd decomposition)**.** *For $1 \le i \le m$:*

(i) $G \in C_{i-1}$ *iff* $(e_G, o_G) \in C_i \times C_i$*, and then* $G(Y) = e_G(Y^2) + Y\, o_G(Y^2)$ *and* $\mathrm{Fold}_r(G) \in C_i$*;*

(ii) $\mathrm{d}(g, g') \le \Delta(g, g')$*;*

(iii) $\mathrm{d}(\mathrm{Fold}_r g, \mathrm{Fold}_r g') \le \Delta(g, g')$*;*

(iv) for $F \in K[Y]_{<L}$*,* $\mathrm{Fold}_{r_m} \circ \cdots \circ \mathrm{Fold}_{r_1}(F|_{D_m})$ *is the constant* $\kappa^{-1}(F)(r_1, \ldots, r_m)$*.*

Proof. (i) $G = E_G(Y^2) + YO_G(Y^2)$ with $\deg E_G, \deg O_G < L_i$, and $e_G = E_G|_{E_i}$, $o_G = O_G|_{E_i}$. (ii) A pair disagreement accounts for at most two point disagreements: $\mathrm{d}(g, g') \le 2\Delta(g, g')n_i/n_{i-1} = \Delta(g, g')$. (iii) The folds agree wherever the pairs agree. (iv) Lemma 3.2 at $X = 1$ (Lemma 8.1(i) with $x = 1$). □

8.2 An elementary line lemma

Lemma 8.5 (Correlated agreement for lines)**.** *Let $C \subseteq K^N$ be linear with minimum distance d, let $t \ge 0$ be an integer with $3t < d$, and let $e, o \in K^N$. Put*

$$R := \{r \in K : \ \exists c \in C, \ |\{x : \ e_x + r o_x \ne c_x\}| \le t\}.$$

If $|R| \ge \max(3, t + 2)$, then there are $E, O \in C$ with $|\{x : \ (e_x, o_x) \ne (E_x, O_x)\}| \le t$.

Proof. For $r \in R$ the codeword c_r is unique since $2t < d$; let A_r be its agreement set, $|A_r| \ge N - t$. Fix distinct $r_1, r_2 \in R$ and set $O := (c_{r_1} - c_{r_2})/(r_1 - r_2)$, $E := c_{r_1} - r_1 O \in C$. On $A_{r_1} \cap A_{r_2}$ the two equations $e + r_j o = c_{r_j}$ give $o = O$, $e = E$. For any $r \in R$, the codewords c_r and $E + rO$ agree on $A_{r_1} \cap A_{r_2} \cap A_r$, of size at least $N - 3t > N - d$, hence $c_r = E + rO$.

Let $D := \{x : (e_x, o_x) \ne (E_x, O_x)\}$. For $x \in D$ the affine function $r \mapsto (e_x - E_x) + r(o_x - O_x)$ is not identically zero, so it vanishes for at most one r; let κ_r be the number of $x \in D$ where it vanishes at r, so $\sum_r \kappa_r \le |D|$. For $r \in R$, the disagreement set of $e + ro$ with $c_r = E + rO$ has size $|D| - \kappa_r \le t$. Summing over R: $|R|(|D| - t) \le |D|$. If $|D| \ge t + 1$ this gives $|R| \le |D|/(|D| - t) \le t + 1$, a contradiction. □

Corollary 8.6. *If e, o are fixed and r is uniform in K, and no $E, O \in C$ satisfy $|\{x : (e_x, o_x) \ne (E_x, O_x)\}| \le t$, then $\Pr_r[r \in R] \le \max(2, t + 1)/|K|$.*

Lemma 8.7 (Correlated agreement for Reed–Solomon lines up to unique decoding)**.** *Let $C = \mathrm{RS}[K, \Omega, k]$ with $|\Omega| = N$ and minimum distance $d = N - k + 1$, let $t \ge 0$ be an integer with $2t < d$, and let $e, o \in K^\Omega$. With R as in Lemma 8.5, if*

$$|R| \ \ge \ (k + 2t)(N + 1) + 2,$$

then there are $E', O' \in C$ with $|\{x \in \Omega : \ (e_x, o_x) \ne (E'(x), O'(x))\}| \le t$.

Proof. The Berlekamp–Welch system over $K[Z]$. Consider the unknowns $\xi_0, \dots, \xi_{t-1}$ and $\zeta_0, \dots, \zeta_{k+t-1}$, the polynomials

$$\Gamma(X) = X^t + \sum_{j<t} \xi_j X^j, \qquad \Psi(X) = \sum_{j<k+t} \zeta_j X^j,$$

and impose, for every $x \in \Omega$,

$$\Psi(x) - \Gamma(x)\,(e_x + Z\,o_x) = 0.$$

This is a linear system $M(Z)\,w = b(Z)$ with N equations and $u = k+2t \le N$ unknowns, whose coefficients are affine in Z. Let $\varrho \le u$ be the rank of $M(Z)$ over the field $K(Z)$.

Step 1: specializations are solvable on R. Let $r \in R$, let $P_r \in C$ be the codeword with at most t disagreements with $w_r := e + ro$ (unique since $2t < d$), and Err_r the disagreement set. Then $\Gamma_r := (X-a)^{t-|\mathrm{Err}_r|} \prod_{x \in \mathrm{Err}_r} (X - x)$ (any $a \in K$) is monic of degree t, $\Psi_r := \Gamma_r P_r$ has degree $< k+t$, and both sides of the equation vanish at $x \in \mathrm{Err}_r$ and agree elsewhere. So $M(r)w = b(r)$ is solvable.

Step 2: the system is solvable over $K(Z)$. Every $(\varrho+1)$-minor of $M(Z)$ is the zero polynomial, so $\operatorname{rank} M(r) \le \varrho$ for all r. If the augmented matrix $[M \,|\, b]$ had rank $\varrho+1$ over $K(Z)$, some $(\varrho+1)$-minor of it would be a nonzero polynomial of degree at most $\varrho+1$; by Step 1 it vanishes at every $r \in R$, contradicting $|R| \ge \varrho+2$. Hence $[M \,|\, b]$ has rank ϱ.

Step 3: a polynomial solution. Choose rows I and columns J with $\Delta(Z) := \det M_{I,J}(Z) \neq 0$, $|I| = |J| = \varrho$. Rows I of $[M \,|\, b]$ are a basis of its row space (Step 2), so the vector with w_J given by Cramer's rule on $M_{I,J} w_J = b_I$ and $w_j = 0$ for $j \notin J$ solves the whole system. Each w_j is $\Delta_j(Z)/\Delta(Z)$ with Δ_j a $\varrho \times \varrho$ minor of a matrix with affine entries, so $\deg \Delta_j \le \varrho$ and $\deg \Delta \le \varrho$. Clearing the denominator,

$$\widetilde{\Gamma}(X,Z) := \Delta(Z)\,\Gamma(X), \qquad \widetilde{\Psi}(X,Z) := \Delta(Z)\,\Psi(X)$$

are polynomials of Z-degree at most ϱ, $\widetilde{\Gamma}$ has X-degree t with leading coefficient $\Delta(Z)$, $\deg_X \widetilde{\Psi} < k+t$, and

$$\widetilde{\Psi}(x,Z) = \widetilde{\Gamma}(x,Z)\,(e_x + Z\,o_x) \qquad \text{in } K[Z], \text{ for every } x \in \Omega. \tag{$*$}$$

Step 4: the common agreement set. Let $S := \{x \in \Omega : \ \widetilde{\Gamma}(x,Z) \neq 0 \text{ in } K[Z]\}$. Each $x \in \Omega \setminus S$ is a root in $K(Z)$ of the degree-t polynomial $\widetilde{\Gamma}(X,Z) \in K(Z)[X]$, so $|\Omega \setminus S| \le t$. Let

$$R'' := \{r \in R : \ \Delta(r) \neq 0 \text{ and } \widetilde{\Gamma}(x,r) \neq 0 \text{ for all } x \in S\}.$$

Δ and each $\widetilde{\Gamma}(x,Z)$, $x \in S$, are nonzero of degree at most ϱ, so $|R \setminus R''| \le \varrho(|S|+1) \le (k+2t)(N+1)$ and $|R''| \ge 2$.

Step 5: unique decoding at each $r \in R''$. Specialize $(*)$ at r: $\widetilde{\Psi}(\cdot, r) = \widetilde{\Gamma}(\cdot, r)\, w_r$ on Ω, with $\deg \widetilde{\Gamma}(\cdot, r) = t$ since $\Delta(r) \neq 0$. The polynomial $\widetilde{\Psi}(\cdot,r) - \widetilde{\Gamma}(\cdot,r) P_r$ has degree $< k+t$ and vanishes on the at least $N-t$ points where $w_r = P_r$; since $k+t \le N-t$ (equivalent to $2t < d$), it is zero. For $x \in S$, $\widetilde{\Gamma}(x,r) \neq 0$, hence

$$e_x + r\,o_x = P_r(x) \qquad (x \in S,\ r \in R'').$$

Step 6: conclusion. Take distinct $r_1, r_2 \in R''$ and put $O' := (P_{r_1} - P_{r_2})/(r_1 - r_2)$, $E' := P_{r_1} - r_1 O'$, both in C. Step 5 at r_1, r_2 gives $o_x = O'(x)$ and $e_x = E'(x)$ for all $x \in S$, and $|\Omega \setminus S| \le t$. □

Corollary 8.8. *In the setting of Lemma 8.7, if e, o are fixed, r is uniform in K, and no $E', O' \in C$ have at most t joint disagreements with (e,o), then* $\Pr_r[r \in R] \le \big((k+2t)(N+1)+1\big)/|K| \le (N+1)^2/|K|$.

8.3 Binding of scalar FRI

Fix a radius δ and set $t_i := \lceil \delta n_i \rceil - 1$, so that "relative distance $< \delta$" and "at most t_i disagreements" coincide on E_i. We use one of two choices:

(RS) $\delta = \theta = \frac{1-\rho}{2}$. Then $2t_i < (1-\rho)n_i < d_i$, Lemma 8.7 applies to every C_i, and we set $\epsilon_i := (n_i+1)^2$.

(Lin) $\delta = \delta^* = \frac{1-\rho}{3}$. Then $3t_i < d_i$, Lemma 8.5 applies to any linear codes with the decomposition of Lemma 8.4, and we set $\epsilon_i := n_i$.

In both cases $2\delta \le 1-\rho < \lambda_i$ for all i. For $B=4$: $\theta = \frac{3}{8}$ and $\delta^* = \frac{1}{4}$.

Definition 8.9 (Reference objects)**.** In a run of Π_1:

- $F_\star$ is the unique $F \in K[Y]_{<L}$ with $\mathrm{d}(s_0, F|_{D_m}) < \delta$, if it exists; $f_\star = \kappa^{-1}(F_\star)$.
- $\varphi_0 := g_0$ and $\varphi_i := \mathrm{Fold}_{r_i}(g_{i-1})$ for $1 \le i \le m$ (the values the verifier recomputes).
- G_j $(0 \le j < m)$ is the unique $G \in C_j$ with $\Delta(g_j, G) < \delta$, if it exists (unique by Lemma 8.4(ii) and $2\delta < \lambda_j$). It is a function of g_j, hence fixed before r_{j+1}.
- For a codeword $H \in C_j$, $H^{\to m} := \mathrm{Fold}_{r_m} \circ \cdots \circ \mathrm{Fold}_{r_{j+1}}(H)$, a constant.

Definition 8.10 (Bad rounds)**.** Round $j+1$ $(0 \le j < m)$ is *bad* if either

(a) G_j does not exist but $\mathrm{d}(\varphi_{j+1}, C_{j+1}) < \delta$; or

(b) for some reference $H \in \{\mathrm{Fold}_{r_j} \circ \cdots \circ \mathrm{Fold}_{r_{i+1}}(G_i) : \ 0 \le i \le j,\ G_i \text{ exists}\} \subseteq C_j$ and some $z \in E_{j+1}$, the pairs of g_j and H differ at z but $\mathrm{Fold}_{r_{j+1}}(g_j)(z) = \mathrm{Fold}_{r_{j+1}}(H)(z)$.

Lemma 8.11. $\Pr[\textit{some round is bad}] \le \dfrac{1}{|K|} \displaystyle\sum_{i=1}^{m} \big(\epsilon_i + i\, n_i\big)$. *In case (RS) this is at most* $(n^2/3 + 4n + m)/|K|$*; in case (Lin) at most* $3n/|K|$.

Proof. All objects in the definition of round $i = j+1$ are fixed before r_i is sampled. Clause (a) has probability at most $\epsilon_i/|K|$ by Corollary 8.8 (case RS) or Corollary 8.6 (case Lin, $\max(2, t_i+1) \le n_i$), applied to $(e_{g_{i-1}}, o_{g_{i-1}})$ and C_i. For clause (b), each of the at most i references and each $z \in E_i$ contribute at most one value of r_i, so at most $i\, n_i/|K|$. Summing with $n_i = n2^{-i}$: $\sum_i (n_i+1)^2 \le n^2/3 + 2n + m$, $\sum_i i\, n_i \le 2n$, and $\sum_i (i+1) n_i \le 3n$. □

From now on fix a run with no bad round in which the final layer g_m is the constant c (otherwise the verifier rejects). For $0 \le i \le m$ and a uniform $z \in E_i$ let $\mathrm{Pass}_i(z)$ be the event that the checks $g_k(w) = \varphi_k(w)$ hold at every node $w = z^{2^{k-i}}$, $i \le k \le m$. Then a query from uniform y_0 passes iff $\mathrm{Pass}_0(y_0)$ (at level 0 the check $g_0 = \varphi_0$ is vacuous).

Lemma 8.12 (Propagation)**.** *Let $H \in C_i$ be G_i or a fold of some $G_{i'}$, $i' \le i$ (or, for $i = m$, any constant), with $H^{\to m} = c$. Then for every $z \in E_i$ with $\varphi_i(z) \ne H(z)$, $\mathrm{Pass}_i(z)$ fails.*

Proof. Downward induction on i. For $i = m$: $\varphi_m(z) \ne H(z) = c = g_m(z)$, so the level-$m$ check fails. For $i < m$: if $\varphi_i(z) \ne g_i(z)$ the level-i check fails. Otherwise $g_i(z) \ne H(z)$, so the pairs of g_i and H differ at $w = z^2$; since round $i+1$ is not bad (clause (b), with H among the references), $\varphi_{i+1}(w) \ne \mathrm{Fold}_{r_{i+1}}(H)(w)$. The codeword $\mathrm{Fold}_{r_{i+1}}(H)$ is again a fold of some $G_{i'}$ with $(\mathrm{Fold}_{r_{i+1}} H)^{\to m} = c$, and the induction hypothesis at level $i+1$ shows that $\mathrm{Pass}_{i+1}(w)$, hence $\mathrm{Pass}_i(z)$, fails. □

Lemma 8.13 (Detection)**.** *For $0 \le i \le m$ let H_i be the unique $H \in C_i$ with $\mathrm{d}(\varphi_i, H) < \delta$, if it exists. If H_i does not exist, or $H_i^{\to m} \ne c$, then $\Pr_z[\mathrm{Pass}_i(z)] \le 1 - \delta$.*

Proof. Downward induction on i.

$i = m$. C_m is the code of constants on D_0 and $\Pr_z[\mathrm{Pass}_m(z)] = \Pr_z[\varphi_m(z) = c] = 1 - \mathrm{d}(\varphi_m, c)$. If H_m does not exist, $\mathrm{d}(\varphi_m, c) \geq \delta$. If $H_m \neq c$, then $\mathrm{d}(\varphi_m, c) \geq \lambda_m - \mathrm{d}(\varphi_m, H_m) > \lambda_m - \delta > \delta$.

$i < m$. Note $\Pr_z[\mathrm{Pass}_i(z)] \leq \Pr_w[\mathrm{Pass}_{i+1}(w)]$, since $w = z^2$ is uniform on E_{i+1} and $\mathrm{Pass}_i(z) \subseteq \mathrm{Pass}_{i+1}(z^2)$.

1. *G_i does not exist.* Since round $i+1$ is not bad (clause (a)), $\mathrm{d}(\varphi_{i+1}, C_{i+1}) \geq \delta$, so H_{i+1} does not exist; by induction $\Pr[\mathrm{Pass}_{i+1}] \leq 1 - \delta$.

2. *G_i exists.* By Lemma 8.4(iii), $\mathrm{d}(\varphi_{i+1}, \mathrm{Fold}_{r_{i+1}} G_i) \leq \Delta(g_i, G_i) < \delta$, so $H_{i+1} = \mathrm{Fold}_{r_{i+1}} G_i$ and $H_{i+1}^{\to m} = G_i^{\to m}$.

   2a. If $G_i^{\to m} \neq c$: by induction $\Pr[\mathrm{Pass}_{i+1}] \leq 1 - \delta$.

   2b. If $G_i^{\to m} = c$: by Lemma 8.12 with $H = G_i$, $\mathrm{Pass}_i(z)$ fails whenever $\varphi_i(z) \neq G_i(z)$, so $\Pr_z[\mathrm{Pass}_i(z)] \leq 1 - \mathrm{d}(\varphi_i, G_i)$. The hypothesis of the lemma says H_i is absent or $H_i^{\to m} \neq c = G_i^{\to m}$, so in either case $H_i \neq G_i$; then $\mathrm{d}(\varphi_i, G_i) \geq \delta$, directly if H_i is absent and by $\mathrm{d}(\varphi_i, G_i) \geq \lambda_i - \mathrm{d}(\varphi_i, H_i) > \lambda_i - \delta > \delta$ otherwise. □

Theorem 8.14 (Evaluation binding of scalar FRI)**.** *For every prover in the FRI part of Π_1, with q independent uniform queries,*

$$\Pr\left[\mathit{accept} \ \wedge \ (F_\star \ \mathit{absent} \ \vee \ c \neq f_\star(r_1, \ldots, r_m))\right] \ \leq \ \varepsilon_{\mathrm{fold}} + (1-\delta)^q,$$

where $\varepsilon_{\mathrm{fold}}$ is the bound of Lemma 8.11; in case (RS), $\varepsilon_{\mathrm{fold}} \leq (n^2/3 + 4n + m)/|K|$ and $\delta = \theta$.

Proof. Outside the event of Lemma 8.11, apply Lemma 8.13 at $i = 0$: here $\varphi_0 = g_0 = s_0$, so $H_0 = F_\star|_{D_m}$ when $F_\star$ exists, and $H_0^{\to m} = f_\star(r)$ by Lemma 8.4(iv). Under the event in brackets each query passes with probability at most $1 - \delta$, independently given the commitments. □

8.4 Soundness of Π_1

Definition 8.15 (Game for Π_1)**.** $\widetilde{P}$ wins if the verifier accepts and either $F_\star$ (radius θ, as in Section 6) does not exist or $H \neq \sum_b f_\star(b)$.

Theorem 8.16 (Soundness of Π_1)**.** *In case (RS), for every unbounded interactive oracle prover,*

$$\Pr[\mathsf{Win}] \ \leq \ \frac{m}{|K|} + \frac{n^2/3 + 4n + m}{|K|} + (1-\theta)^q, \qquad \theta = \frac{1-\rho}{2}.$$

Proof. Outside the event of Theorem 8.14, $F_\star$ exists and $c_m = c = f_\star(r)$. The sumcheck messages (u_i, v_i) are sent before r_i, and the honest values $P_i^\star(1) = A_i^\star(r_1, \ldots, r_i)$ depend on $f_\star$, which is fixed before r_1; Lemma 6.3 with the single path through $x = 1$ (one pair $(y, x) = (1, 1)$ per round) bounds the event $H \neq H_\star$ by $m/|K|$. □

Parameters. For Goldilocks^3, $m = 20$, $B = 4$: $\theta = \frac{3}{8}$ and $(5/8)^{189} = 2^{-128.16}$ as for Π_{dual}; $(n^2/3 + 4n + m)/|K| \approx 2^{-149.6}$ and $m/|K| \approx 2^{-187.7}$. Hence $q = 189$ queries give interactive error below 2^{-128}. In case (Lin), the same proof gives $m/|K| + 3n/|K| + (3/4)^q$ for any foldable linear codes, i.e. $q = 309$ at $B = 4$.

Cost of Π_1. $2m$ scalars of sumcheck, and per query one pair and one Merkle path per level. With pairs grouped in leaves and 32-byte digests this is $\sum_{i=0}^{m-1} \big(48 + 32(m - i + 1)\big) = 8320$ bytes per query, about 1.57 MB for $q = 189$ before path deduplication.

Remark 8.17 (Where the Reed–Solomon structure enters)**.** Only clause (a) of the bad-round event uses the code beyond linearity: Lemma 8.7 is the Berlekamp–Welch argument run over $K(Z)$, and it is what lifts the radius from $(1-\rho)/3$ to $(1-\rho)/2$. Its error $(N+1)^2/|K|$ is quadratic in the domain size; for $|K| \approx 2^{192}$ this costs nothing at $m = 20$.

9 Numerical validation

The crate `rust/` of the repository implements Π_{dual} over Goldilocks[3] with $K = \mathbb{F}_p[x]/(x^3 - 2)$ (irreducible, since $2^{(p-1)/3} \neq 1$), the provers of Section 5, and a chain-B tightness prover: s_0 equals F' outside a negation-closed set of relative size $3/8$ where it equals an unrelated codeword, and the higher levels are the honest butterfly of F'. Then $\mathrm{d}(s_0, \mathrm{RS}) = \theta$, so $F_\star$ does not exist and every acceptance is a win. The tests check Lemmas 3.1, 3.2, 2.1, Proposition 4.1 and Remark 1.1 for $m \leq 6$ as exact identities.

Measured acceptance at $m = 5$, $B = 4$, $n = 128$, over 2000 independent runs (fresh challenges and queries):

Prover	measured	predicted
honest, (F1) on	1.0000	1
fiber splitting, (F1) off	1.0000	1
fiber splitting, (F1) on	0.0000	0
late (U_1, V_1), 1 chain-A path	0.4920	0.5
late (U_1, V_1), 2 paths	0.2590	0.25
late (U_1, V_1), 4 paths	0.0720	0.0625
tightness, $q = 1$	0.6325	0.6250
tightness, $q = 2$	0.4020	0.3906
tightness, $q = 4$	0.1595	0.1526
tightness, $q = 8$	0.0260	0.0233

All deviations are within 1.8 standard deviations $\sqrt{p(1-p)/2000}$. The tightness rows show that the bound $(1-\theta)^q$ of Lemma 6.5 is attained.

10 Summary of the security relation

	Π_{dual} (formal X)	Π_1 ($X := 1$)
Chain A	pathwise sumcheck, one path, $\leq 2^{-169}$	classical sumcheck, $m/\lvert K\rvert$
Chain B binding	injective fold, deterministic, $(1-\theta)^q$, $\theta = \frac{1-\rho}{2}$	RS line lemma, $\frac{n^2/3+4n+m}{\lvert K\rvert}+(1-\theta)^q$
Role of r in chain B	none	essential
Terminal link	B full leaves equal, then $P_m(x_m) = T(x_m)$	final layer constant $= c_m$
Queries (128 bits, $B = 4$)	189	189
Proof size ($m = 20$)	≈ 0.85 GB	≈ 1.6 MB
Verifier	$\Theta(L)$	$O(q\,m)$

The dual-chain design separates into a part that needs no proximity theory but is not succinct (the formal butterfly, bound by injectivity) and a succinct part bound by a correlated-agreement lemma for lines (scalar FRI); both reach the unique-decoding radius $(1-\rho)/2$ and the same 189 queries. Remark 1.1 shows that, for a single multilinear polynomial, the sum is the evaluation $2^m f(\frac{1}{2}, \ldots, \frac{1}{2})$; the sumcheck layer becomes necessary once the summand is a product (for instance $f \cdot g$ or $f \cdot \mathrm{eq}$), which is the natural next target for this construction.

References

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